Customer Behavior Analysis for Campaign Optimization

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2024-11-12

Introduction

The **Marketing Campaign Performance Dataset** offers an in-depth look at the performance and impact of various marketing campaigns over two years, providing a solid foundation for analyzing campaign success and customer segmentation. This analysis will leverage data on company campaigns, customer engagement, ROI, and demographics to uncover trends and patterns that can inform future strategies.

Dataset Overview

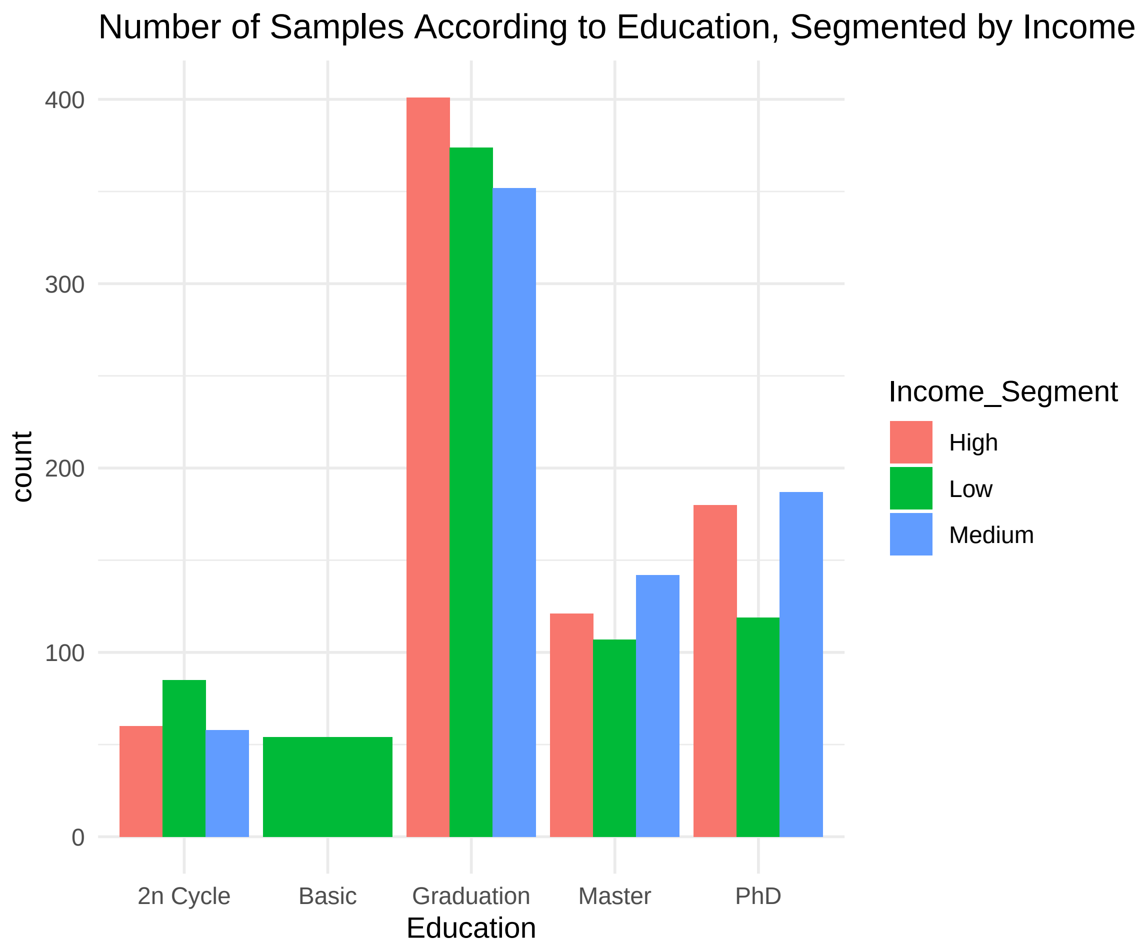
This dataset includes 200,000 entries, each representing a unique marketing campaign. Key columns include:

- **Company**: The fictional brand conducting the campaign.
- **Campaign_Type**: Type of campaign (e.g., email, social media, influencer).
- **Target_Audience**: Demographic targeted, such as specific age and gender groups.
- **Duration**: Length of the campaign in days.
- **Channels_Used**: Platforms where the campaign was promoted.
- **Conversion_Rate**: Percentage of impressions that converted to desired actions.
- **Acquisition_Cost**: Cost incurred for customer acquisition.
- **ROI**: Campaign profitability.
- **Location**: Cities where campaigns ran.
- **Language**: Language used in the campaign communication.
- **Clicks**: User engagement via clicks.
- **Impressions**: Total number of views.
- **Engagement_Score**: Engagement score on a 1–10 scale.
- **Customer_Segment**: Category of the targeted audience.
- **Date**: Campaign date, enabling trend analysis.

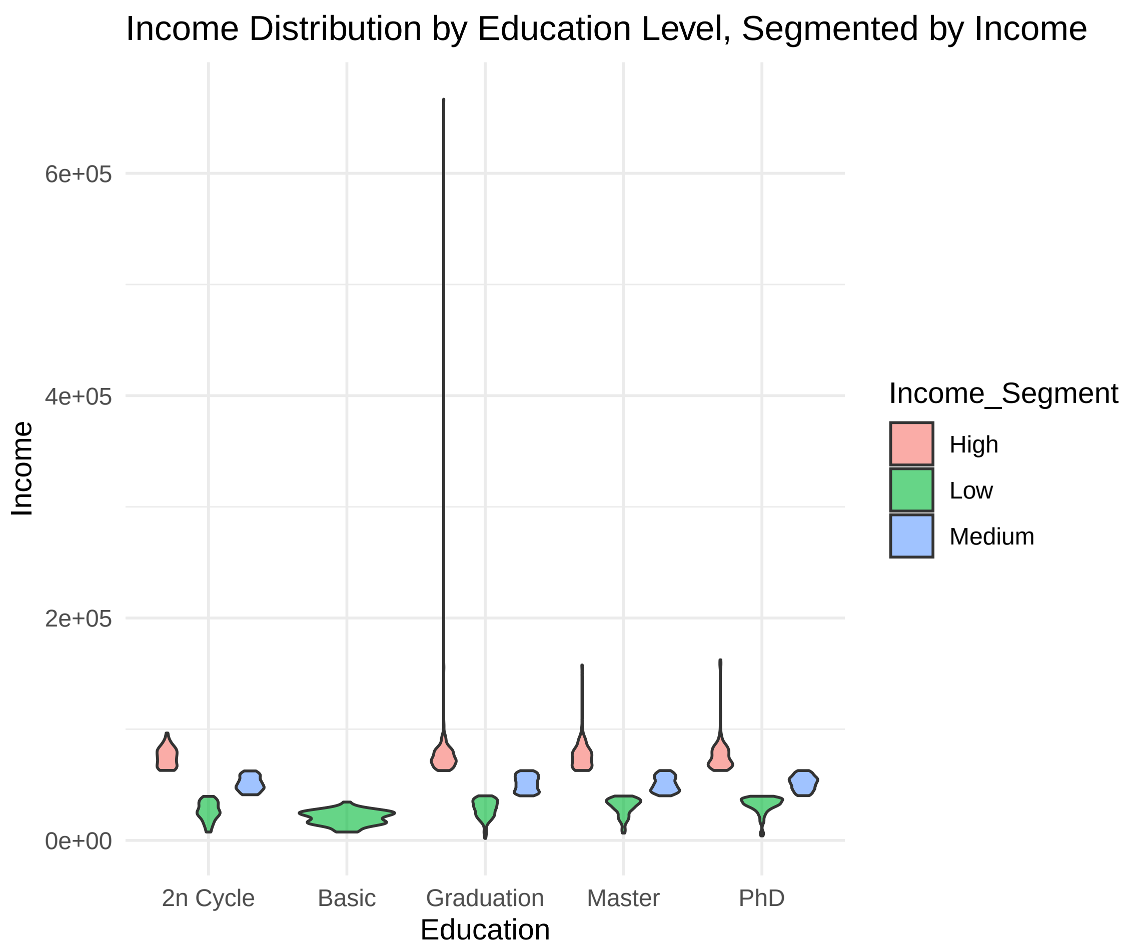
With these variables, the dataset provides rich insights into audience engagement, campaign efficiency, and performance variations across customer segments and geographic areas. In the following sections, we will preprocess and analyze this data through clustering, PCA, and visualization, extracting patterns related to income segmentation, educational levels, and spending behavior.

We are dividing the income data into three distinct segments—Low, Medium, and High—to better understand the behaviors and characteristics of individuals across different income levels. This segmentation enables us to analyze the dataset more effectively by grouping individuals based on their purchasing power.

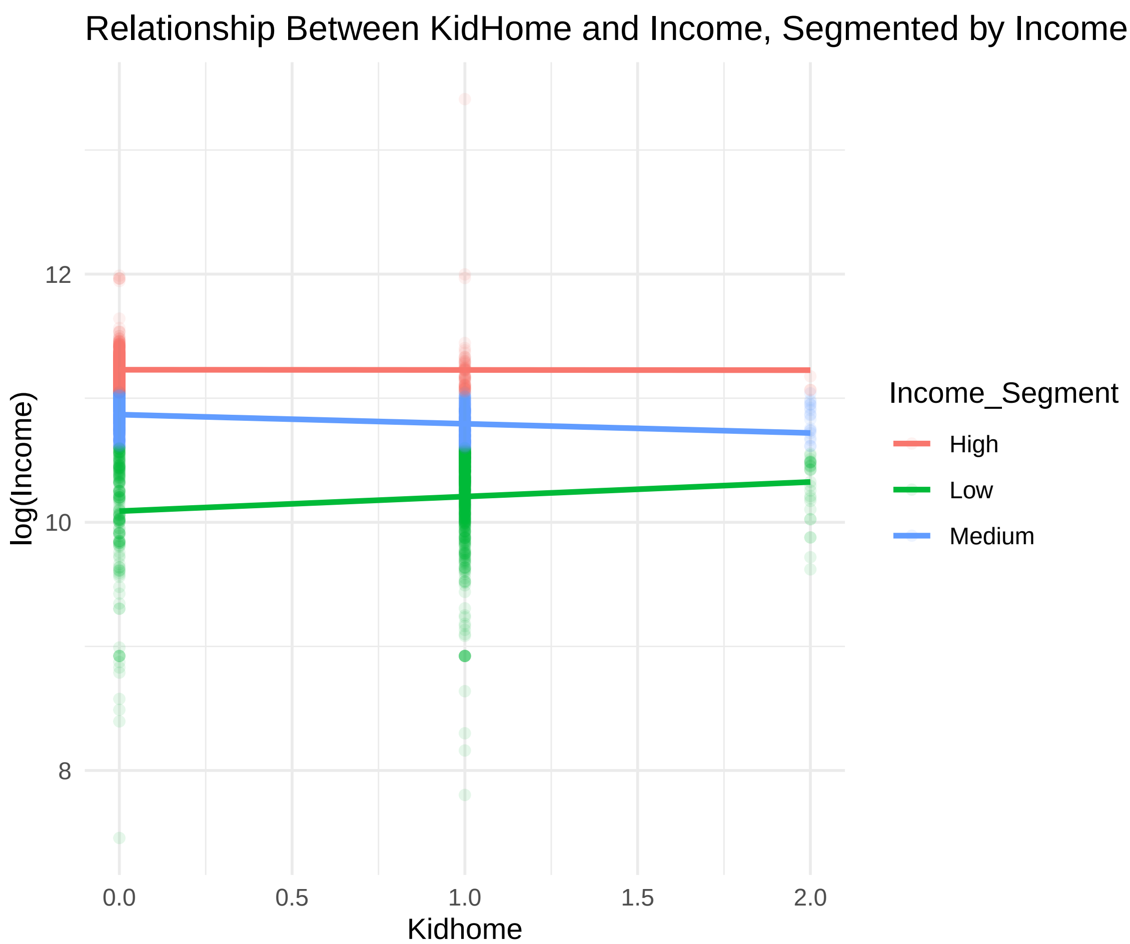
Exploratory Data Analysis (EDA)



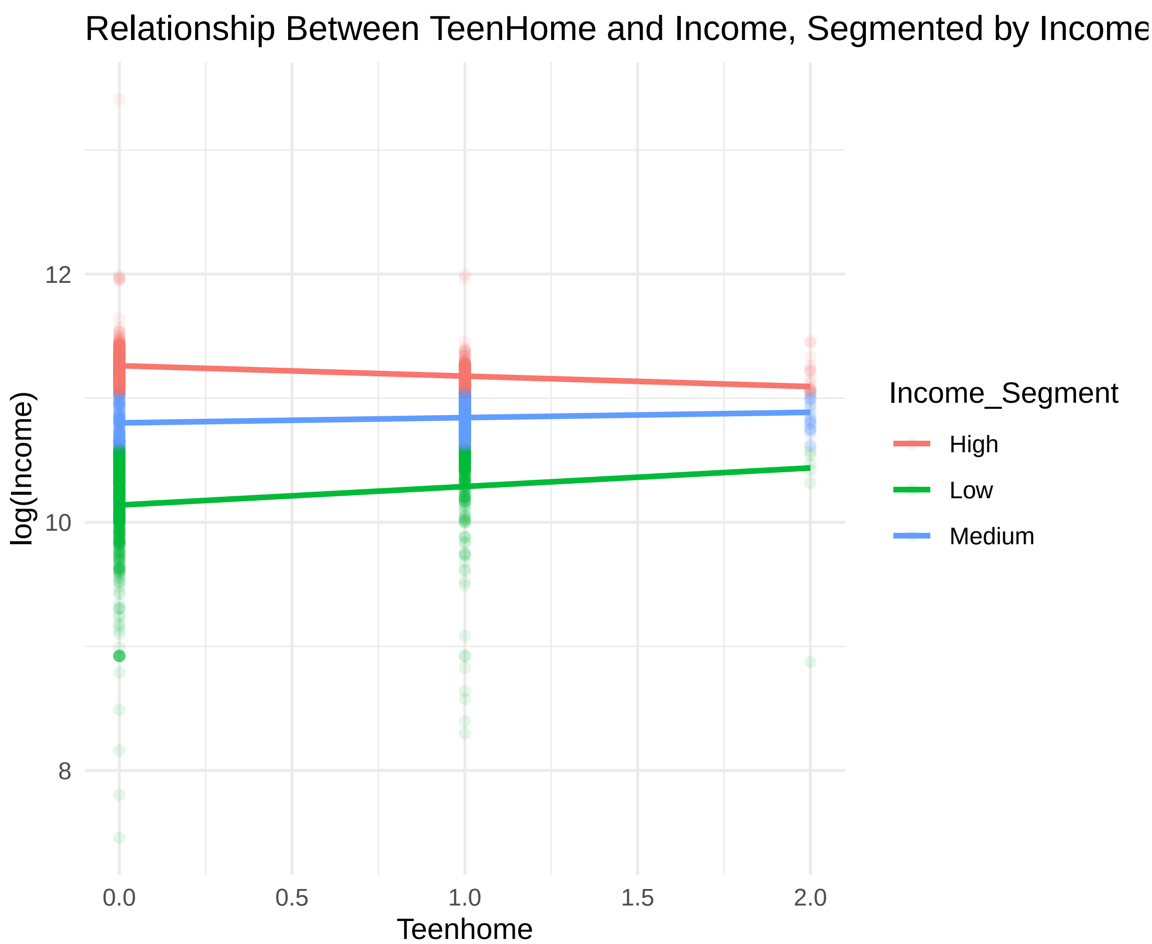
- The highest count of individuals appears within a specific education level, with the "High" income segment (red) as the largest, followed by "Medium" (blue) and "Low" (green).
- The "Medium" income segment has significant representation across most education levels, indicating a broad spread of medium-income individuals.
- Lower education levels have fewer individuals across all income segments, suggesting a smaller representation in these categories.
- Among the lower education levels, the "Low" income segment (green) is relatively higher, indicating a tendency toward lower income within this group.




```
## `geom_smooth()` using formula = 'y ~ x'
```

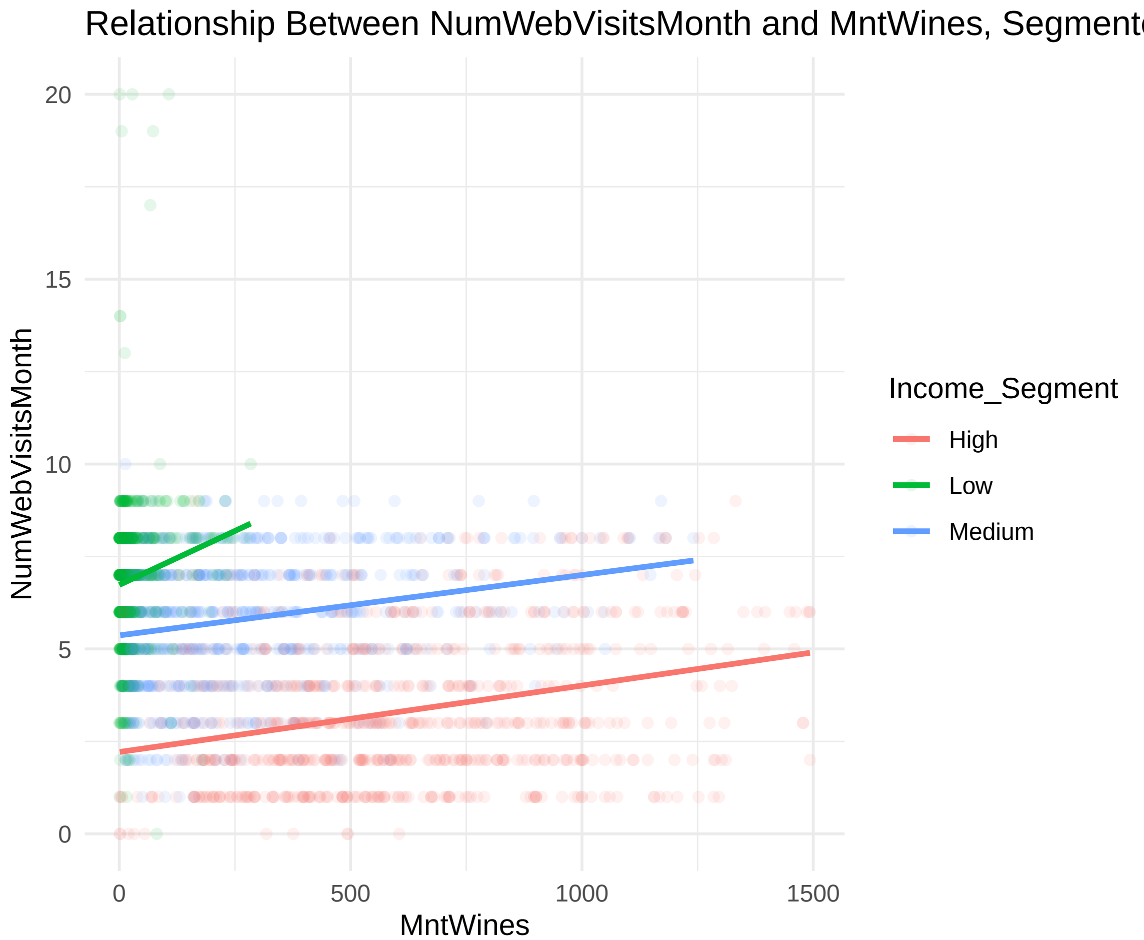



```
## `geom_smooth()` using formula = 'y ~ x'
```



- Income levels remain stable across different numbers of children (`KidHome`) and teenagers (`TeenHome`) at home.
- A clear income hierarchy is maintained, with "High" income consistently highest, followed by "Medium" and "Low," regardless of family size.
- The impact of family composition on income is minimal, showing similar patterns for both `KidHome` and `TeenHome`.

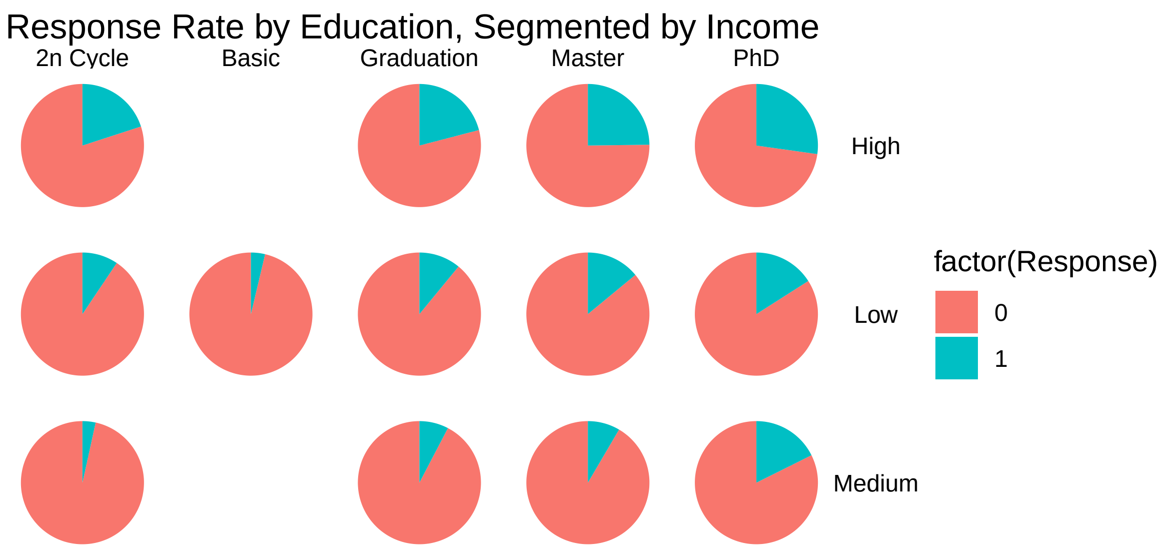
```
## `geom_smooth()` using formula = 'y ~ x'
```



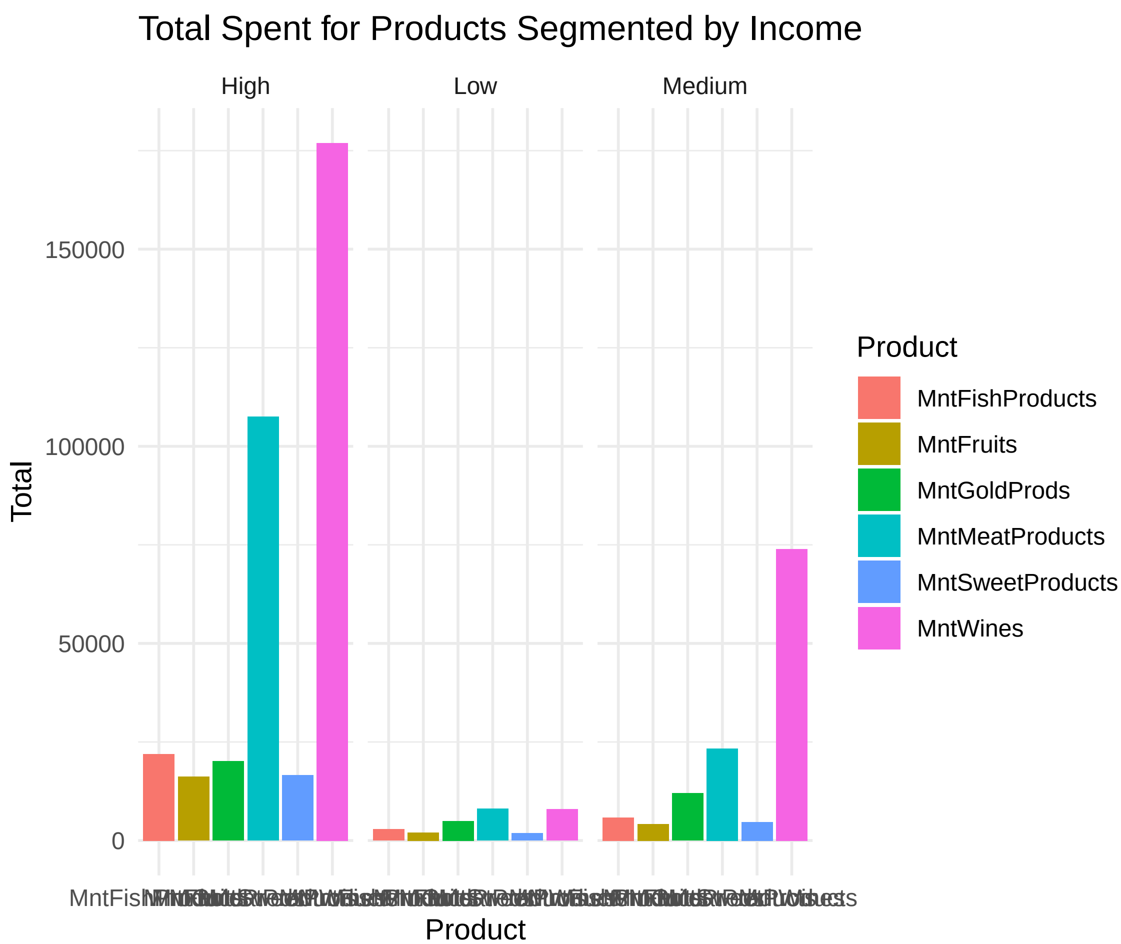
- **Positive Correlation**: There is a slight positive trend between the amount spent on wine (`MntWines`) and the number of website visits per month (`NumWebVisitsMonth`) across all income segments.
- **Income-Based Patterns**: The "High" income segment (red) generally has fewer website visits per month compared to the "Medium" (blue) and "Low" (green) segments, even at higher wine expenditure levels.
- **Distinct Segmentation**: The "Low" income segment shows higher website visits for lower wine spending, while the "Medium" and "High" segments show a more gradual increase in web visits as spending on wine increases.

```
## `geom_smooth()` using formula = 'y ~ x'
```

- **Income-Based Purchase Patterns**: The number of store purchases (`NumStorePurchases`) varies significantly by income segment, with the "Low" income group (green) having fewer purchases at lower income levels, and purchases increasing as income rises.
- **Decline in Purchases for High Income**: In the "High" income segment (red), store purchases decline at higher income levels, indicating that higher-income individuals might prefer other shopping channels or purchase less frequently in-store.
- **Medium Income Segment Stability**: The "Medium" income group (blue) shows a stable and moderate number of store purchases across the income range, suggesting a consistent in-store shopping behavior among medium-income individuals.

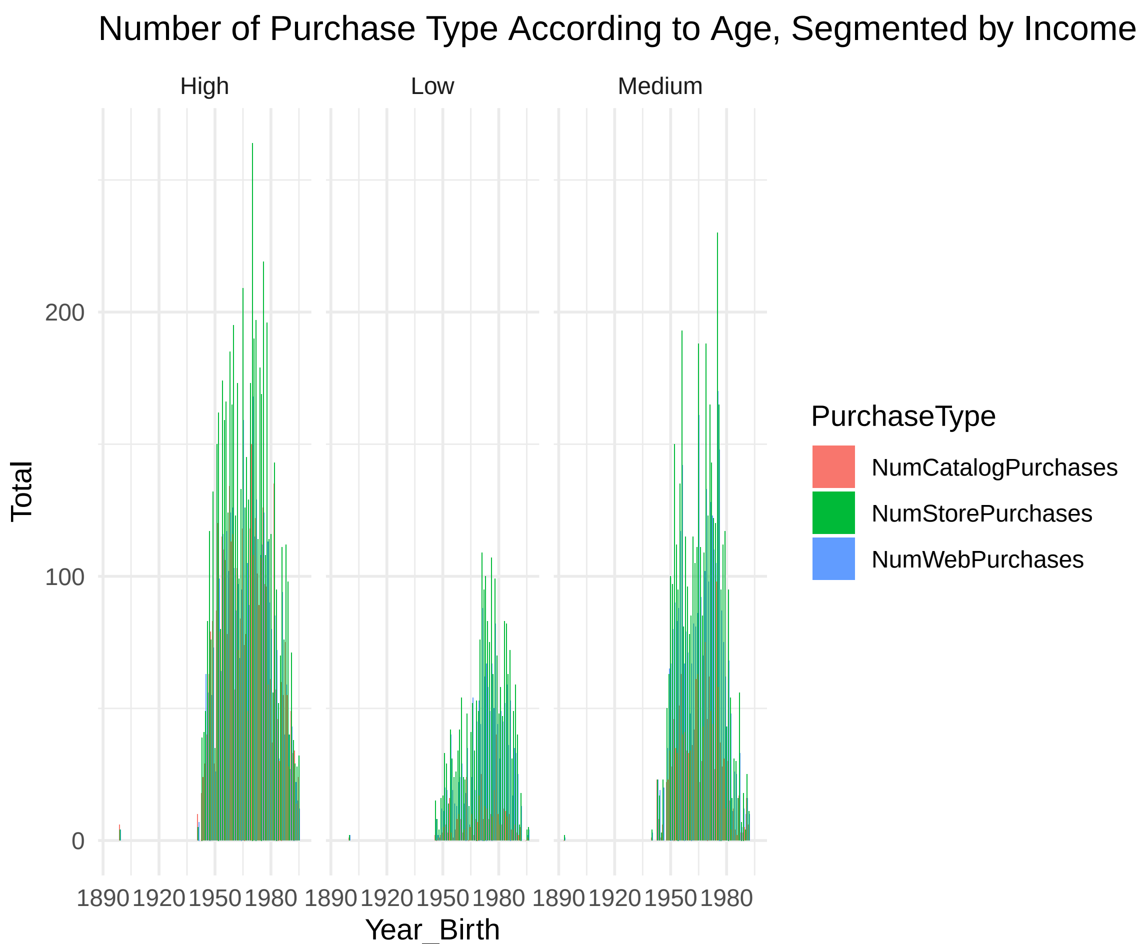



```
## Warning: There was 1 warning in `summarize()`.
## i In argument: `across(starts_with("Mnt"), sum, na.rm = TRUE)`.
## i In group 1: `Marital_Status = "Absurd"` and `Income_Segment = "High"`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))

## `summarise()` has grouped output by 'Marital_Status'. You can override using
## the `.groups` argument.
```



All three income segments exhibit a similar purchasing distribution across product categories. However, the low-income segment spends significantly less on luxury products, such as `MntGoldProds`, compared to the medium- and high-income segments.

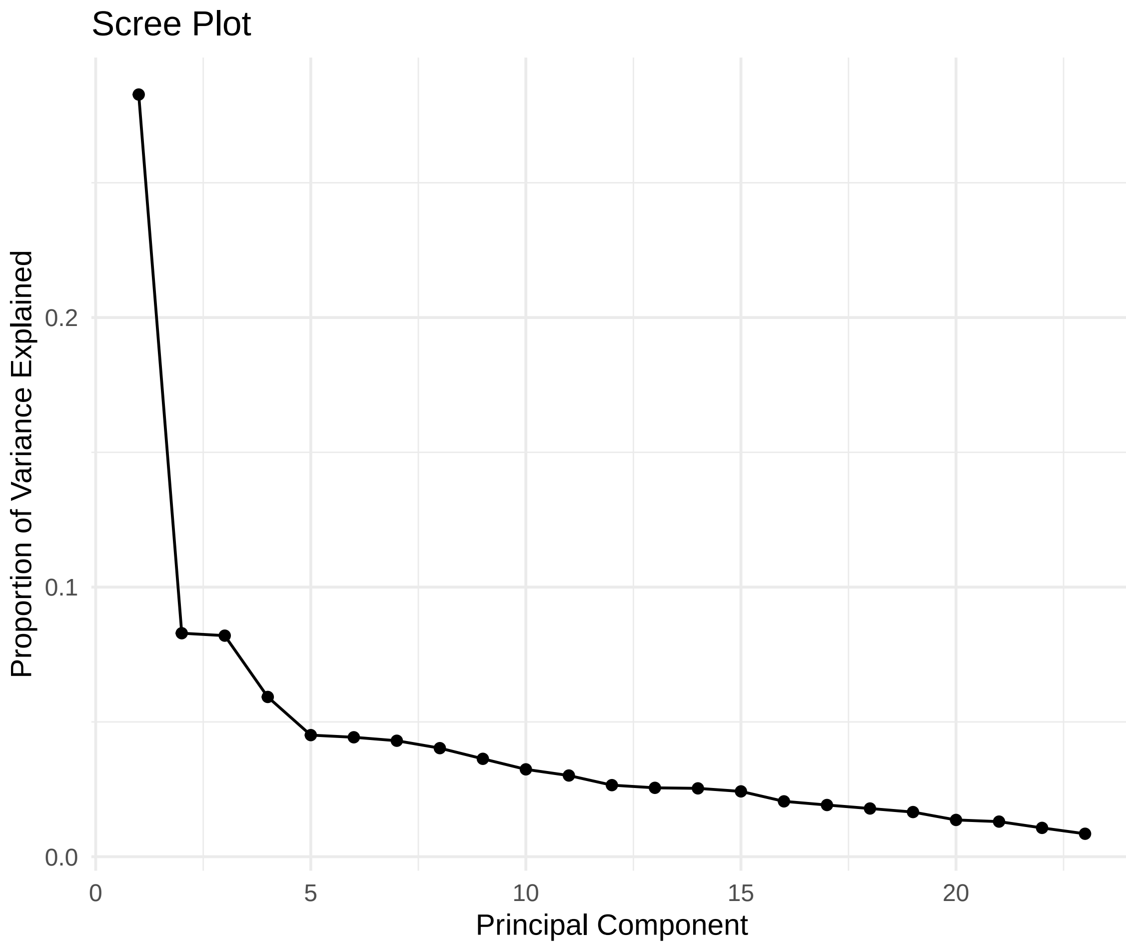
```
## `summarise()` has grouped output by 'Year_Birth'. You can override using the
## `.groups` argument.
```



1. **Similar Purchasing Patterns Across Income Segments**:
    - All three income segments (High, Medium, Low) show a similar distribution of purchase types (catalog, store, and web purchases) across different birth years.
2. **Age-Related Trends**:
    - Younger individuals (born after 1970) across all income segments tend to have more web and store purchases, while older individuals show fewer total purchases, especially in the low-income segment.
3. **Lower Engagement in Low-Income Segment**:
    - The low-income group shows fewer total purchases across all types, particularly for catalog purchases, indicating lower overall engagement or purchasing power in this segment.

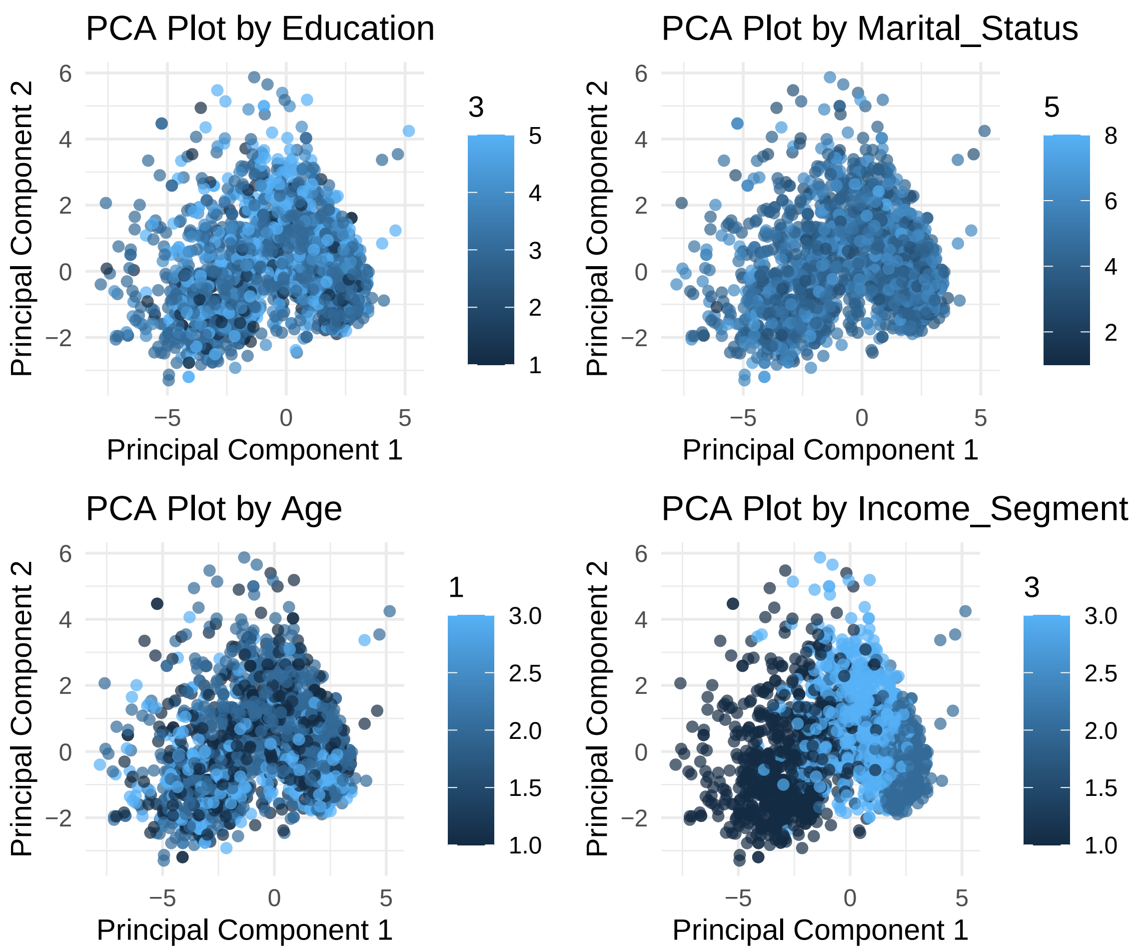
```
##
##  iter imp variable
##   1   1  Income
##   2   1  Income
##   3   1  Income
##   4   1  Income
##   5   1  Income

## Warning: Number of logged events: 5
```

PCA Inference



- The first principal component explains the most variance, followed by a sharp decline in subsequent components.
- An "elbow" is observed around the third component, suggesting that the first three components capture most of the important variance.
- Additional components contribute minimal variance, indicating they may have limited value for representing the data.

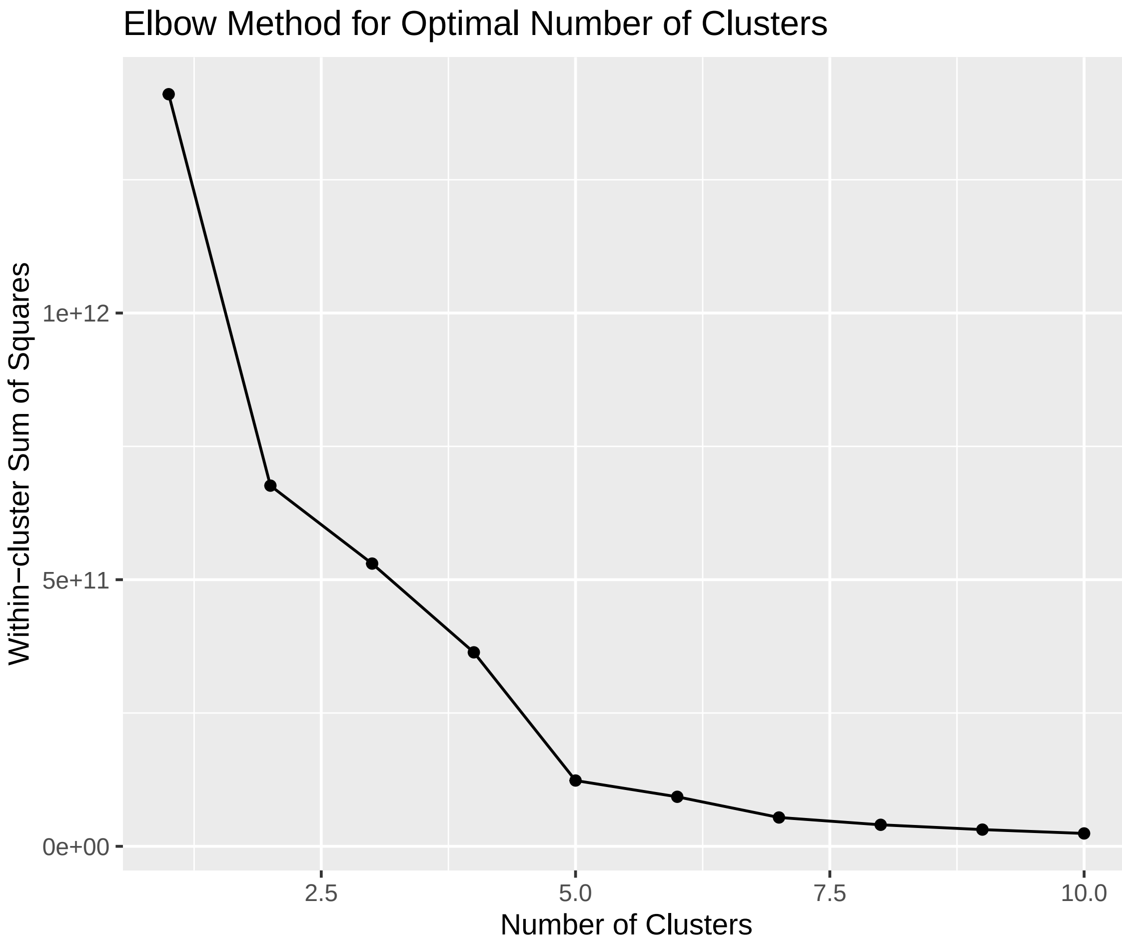
PCA Plots

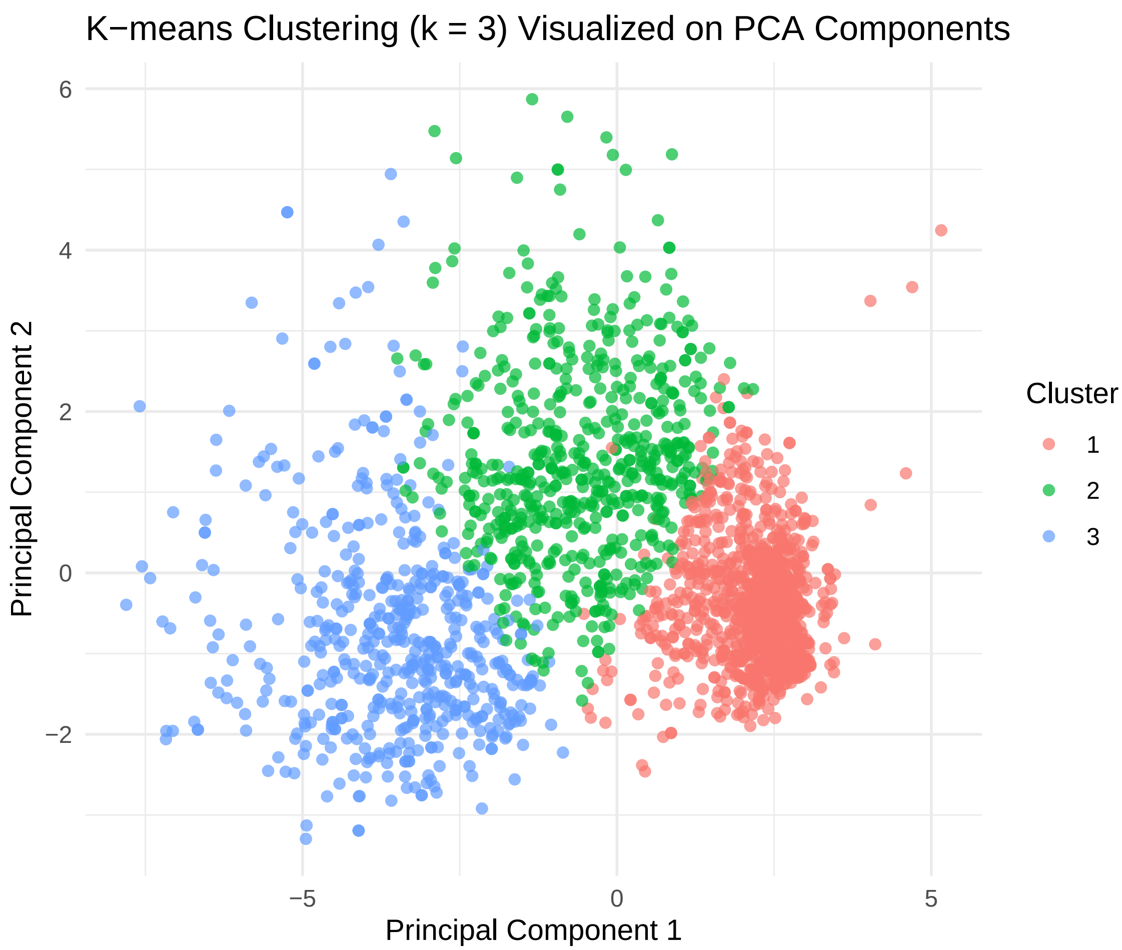


- The PCA plots for **Education**, **Marital Status**, and **Age** show similar clustering patterns, indicating minimal separation among these categories in the principal component space.
- The **Income Segment** plot displays a more noticeable separation, with some clustering based on income level, suggesting that income may have a stronger relationship with the principal components compared to the other variables.
- This indicates that while **Income Segment** may influence the principal components to some extent, **Education**, **Marital Status**, and **Age** do not appear to contribute significantly to variance in this PCA representation.

Unsupervised Clustering

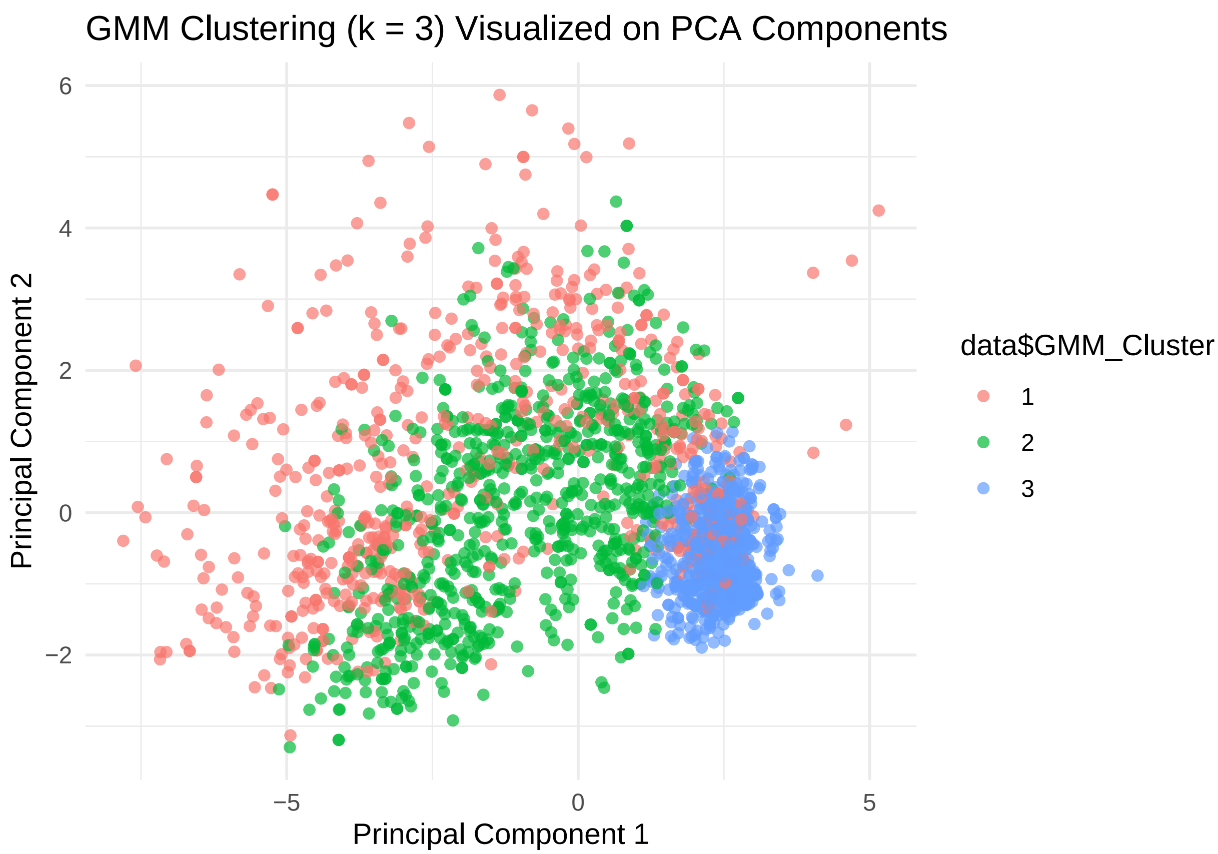
K-means Clustering: K-means is a popular unsupervised clustering algorithm that partitions data into a pre-defined number of clusters (k). It works by initializing k centroids, then iteratively assigning data points to the nearest centroid and updating the centroid positions until convergence. The goal is to minimize the variance within clusters, resulting in compact, spherical clusters. K-means is fast and easy to interpret but assumes clusters of similar size and shape.


Elbow Method for Optimal Number of Clusters
Within−cluster Sum of Squares
1e+12
5e+11
0e+00
2.5
5.0
7.5
10.0
Number of Clusters



Gaussian Mixture Model (GMM): GMM is a probabilistic clustering method that models the data as a mixture of several Gaussian distributions. Unlike K-means, GMM allows for clusters of different shapes, sizes, and orientations. Each data point is assigned a probability of belonging to each cluster, rather than a hard assignment. GMM uses the Expectation-Maximization (EM) algorithm to iteratively optimize the parameters. It is more flexible than K-means but can be more complex and computationally intensive.

```
ggplot(pca_scores, aes(x = PC1, y = PC2, color = data$GMM_Cluster)) +
  geom_point(alpha = 0.7) +
  labs(title = "GMM Clustering (k = 3) Visualized on PCA Components", x = "Principal Component 1", y =
  theme_minimal()
```


GMM Clustering (k = 3) Visualized on PCA Components
Principal Component 2
Principal Component 1
6
4
2
0
−2
−5
0
5
data$GMM_Cluster
1
2
3

kmeans results

```r
contingency_tables <- lapply(columns_to_convert, function(var) table(data[[var]], data$Cluster))
names(contingency_tables) <- columns_to_convert
contingency_tables
```

```
## $Education
##
##       1   2   3
##   1 112  36  55
##   2  52   2   0
##   3 521 301 305
##   4 179 111  80
##   5 201 170 115
##
## $Marital_Status
##
##       1   2   3
##   1   0   0   2
##   2   2   1   0
##   3 104  77  51
##   4 412 253 199
##   5 244 103 133
##   6 276 159 145
##   7  27  25  25
##   8   0   2   0
##
## $Age
##
##       1   2   3
##   1 192 202 176
##   2 542 355 219
##   3 331  63 160
##
## $Income_Segment
##
##       1   2   3
##   1  16 223 523
##   2 711  27   1
##   3 338 370  31
```

gmm results

```
contingency_tables <- lapply(columns_to_convert, function(var) table(data[[var]], data$GMM_Cluster))
names(contingency_tables) <- columns_to_convert
contingency_tables
```

```
## $Education
##
##       1   2   3
##   1  52  61  90
##   2   8   3  43
##   3 312 436 379
##   4 103 149 118
##   5 170 184 132
##
## $Marital_Status
##
##       1   2   3
##   1   1   1   0
##   2   1   1   1
##   3  71  86  75
##   4 235 325 304
##   5 156 151 173
##   6 152 238 190
##   7  28  30  19
##   8   1   1   0
##
## $Age
##
##       1   2   3
##   1 176 274 120
##   2 303 420 393
##   3 166 139 249
##
## $Income_Segment
##
##       1   2   3
##   1 345 415   2
##   2 130  43 566
##   3 170 375 194
```

By these results we can verify that by simply clustering we arent able to get any of the segement properly.

Discriminant Analysis and Bayesian Classification

Linear Discriminant Analysis (LDA): LDA is a classification technique that assumes classes have a Gaussian distribution with the same covariance matrix but different means. It projects data onto a lower-dimensional space where the separation between classes is maximized. LDA is effective when class distributions are approximately linear and is often used in cases where the goal is to reduce dimensionality while preserving class separability.

Quadratic Discriminant Analysis (QDA): QDA is similar to LDA but allows each class to have its own covariance matrix, which enables it to capture non-linear decision boundaries between classes. This makes

QDA more flexible than LDA, particularly for complex datasets, but it requires more data to estimate the individual covariance matrices accurately.

Naive Bayes Classifier: The Naive Bayes classifier is a probabilistic model based on Bayes' theorem, assuming that features are independent given the class label (hence "naive"). It calculates the probability of each class given the input features and assigns the class with the highest probability. Naive Bayes is simple, fast, and effective, especially for text classification, but its independence assumption may limit its performance on datasets with correlated features.

Class_Variable	Model	Misclassification_Rate
Education	LDA	0.4439462
Education	QDA	NA
Education	Naive Bayes	0.7713004
Marital_Status	LDA	0.6457399
Marital_Status	QDA	NA
Marital_Status	Naive Bayes	0.9730942
Age	LDA	0.5022422
Age	QDA	0.5022422
Age	Naive Bayes	0.5201794
Year	LDA	0.3811659
Year	QDA	0.3946188
Year	Naive Bayes	0.4349776
Income_Segment	LDA	0.1614350
Income_Segment	QDA	NA
Income_Segment	Naive Bayes	0.2780269

LDA works best on this dataset as it captures the main patterns and provides clear decision boundaries between classes.

Future Work

- We plan to apply **association rule mining** to uncover hidden patterns and relationships within the dataset, which can provide valuable insights into frequently co-occurring features and behaviors.
- We intend to explore various **neural network architectures** and other advanced machine learning models to evaluate whether they can achieve better classification results than traditional methods.
- We will conduct a comparative analysis across multiple models, examining not only their accuracy but also the correlation in their predictions, to assess whether complex models align with the simpler models currently in use.